

Yangon Tech Hub Project Rules & Coding Standards

This document defines the official development rules, coding standards, workflow process, and quality requirements for the Smart ERP & Workforce Management Platform project developed for Yangon Tech Hub.

1. Development Workflow

- All tasks must be tracked through the project management board.
- Developers must create feature branches before implementation.
- Pull Requests (PR) are required before merging into main branch.
- Code review approval is mandatory before deployment.
- Daily progress updates are required from all team members.

2. Frontend Standards

- Frontend framework must use Next.js.
- Reusable components should be created whenever possible.
- Use TypeScript for all frontend development.
- Responsive design is required for desktop and mobile screens.
- Avoid inline styles; use modular or utility-based styling.

3. Backend Standards

- Backend services must use Spring Boot REST APIs.
- Use layered architecture (Controller, Service, Repository).
- All APIs must include validation and error handling.
- JWT authentication and role-based access control are mandatory.
- API documentation must be maintained using Swagger/OpenAPI.

4. Database Rules

- MySQL is the official project database.
- All tables must include created_at and updated_at timestamps.
- Use foreign keys and indexes properly.
- Database migration scripts must be version controlled.
- Sensitive information must never be stored as plain text.

5. Git & Branch Naming

- feature/* for new features
- bugfix/* for fixes
- hotfix/* for urgent production fixes
- release/* for release preparation
- Commit messages must be clear and meaningful.

6. Testing & QA

- Unit testing is required for critical business logic.
- QA testing must be completed before release.
- All bugs must be documented and tracked.
- Regression testing is required before production deployment.

7. Security Rules

- Passwords must be encrypted.
- Role-based permissions are mandatory.
- All APIs must validate authentication tokens.
- Prevent SQL Injection and XSS vulnerabilities.
- Regular backup procedures must be maintained.

8. Deployment Rules

- Production deployments require PM approval.
- Environment variables must not be hardcoded.
- CI/CD pipeline should be used for deployment automation.
- Production logs must be monitored regularly.

Approved By: Project Manager 1
Client: Yangon Tech Hub
Status: Active Development